

Discrete Random Expected Utility and Self-Selection

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Abstract

We model stochastic choice rules via finitely many types θ that maximize distinct expected utility functions and use endogenous tie-breaking rules. First, we characterize *discrete random expected utility* (DREU) where the likelihood $\mu(\theta)$ of each relevant type θ is preserved across all menus A . This model is a discrete version for the *random expected utility* of Gul and Pesendorfer (2006), but our axioms, identification, and tie-breaking procedures are novel. More generally, we propose *discrete-map expected utility* (DMEU), where the likelihoods $\mu_A(\theta)$ are contingent on the menu A . The continuous map μ_A is identified uniquely in our model. DMEU captures various kinds of *context dependence*, such as reason-based choice, extremeness aversion, and other behavioral patterns. Moreover, we use DMEU to model *self-selection*, where types can increase their participation rates across distinct menus, but only if their best choices are improved. The standard *monotonicity* principle for stochastic choices delivers a novel *self-selective* property for the type likelihoods μ_A . This bias is identifiable because all types are assumed to maximize expected utility functions. By contrast, it can distort the likelihoods $\mu(\theta)$ without turning Block-Marschak polynomials negative in the general random utility model. Finally, we discuss applications to random *risk aversion* and random *Cobb-Douglas utility*.

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1 Introduction

Observed choices in consumption menus can be naturally *stochastic* when they are produced by distinct rational (i.e., utility maximizing) agents. It is common for data to be presented as an aggregation of choices from diverse agents. Such data is used in a wide range of empirical work, such as preferred modes of transportation (McFadden, 2001) and choice of fishing sites (Train, 1998). This heterogeneity may also result from more behavioral motivations where a single individual can exhibit distinct random responses in identical choice problems (see Agranov and Ortoleva (2017)).

The classic random utility model (RUM) by Block and Marschak (1960) aims to represent stochastic choice data by a stable probability distribution μ across *types* that include all possible utility functions. In the general case where types are unrestricted, Falmagne (1978) characterizes RUM via non-negativity of Block and Marschak’s polynomials. Gul and Penderfer (2006) (henceforth, GP) refine this model to *random expected utility* (REU) where all types should maximize expected utility functions rather than arbitrary ones. Other refinements of RUM have restricted types by single-crossing conditions (Apesteguia, Ballester, and Lu, 2017), quasi-linearity (McFadden, 1973; Williams, 1977; Daly and Zachery, 1979; Yang and Kopylov, 2023), and other assumptions. Behavioral patterns, such as ambiguity aversion (Lu, 2021), have also been discussed within the context of random utility. All of these refinements preserve the assumption that μ is invariant of the consumption menu.

In this paper, we propose a novel version of random utility that has

- a discrete (i.e., finite) type space Θ where all types $\theta \in \Theta$ combine expected utility maximization with endogenous tie-breaking rules;
- a distribution of types μ_A that can depend on the menu A and capture context dependence and self-selection;
- a unique identification for both Θ and μ_A .

We still consider a representation where the distribution $\mu(\theta)$ is invariant of A . This case is a discrete version of the GP’s REU, but our axioms and identifications are more parsimonious than theirs and rely more on algebraic rather than analytical techniques. Moreover, the algebraic approach makes the consumption domain X more flexible than in GP, where X is a finite-dimensional simplex of lotteries. In our framework, X is convex, but need not be closed, or bounded, or finite dimensional. For example, X may consist of monetary gambles with payoffs on the real line, or X can be the non-negative orthant of consumption bundles.

More substantially, we depart from REU by allowing the distribution μ_A to vary with the menu A . In general, it is only required that the support $\Theta = \{\theta : \mu_A(\theta) > 0\}$ be finite

and invariant of A . This Θ is called a discrete type space. It is used to represent observable likelihoods $\rho(x, A)$ of choosing any element x in any finite consumption menu A . Our most general representation takes the form

$$\rho(x, A) = \mu_A\{\theta \in \Theta : x \text{ is the best element for } \theta \text{ in } A\} \quad (1)$$

where $\mu_A(\theta)$ is the probability of type θ making a choice in the menu A .

Each type θ is assumed to maximize some expected utility function u_θ in each menu A . However, in order for representation (1) to be well-defined, each type $\theta \in \Theta$ must also incorporate some endogenous tie-breaking rule that selects a unique maximizer for u_θ in A . Formally, each θ is associated with a total, complete, transitive binary relation that satisfies von Neumann-Morgenstern's Independence axiom as well. Such total types have been used in finite-dimensional contexts by Fishburn (1982), Myerson (1986), Blume, Branderburger, and Dekel (1991) and others. We argue that there is a rich family of total types even if X is not finite dimensional. Thus we provide a novel way to accommodate tie-breaking in random expected utility. Without totality, Piermont (2022) relaxes GP's model to represent only those choices that are strict maximizers for corresponding types.

We show several characterization results for stochastic choice rules. First, Theorem 2 below characterizes representation (1) where $\mu_A = \mu$ is menu-invariant and relates this result to GP's model. Our main result (Theorem 3 below) characterizes representation (1) and provides a convenient platform for various refinements that impose more structure on the menu-contingent distribution μ_A and/or types θ . We obtain several such refinements.

Theorem 4 summarizes the implications of the standard monotonicity principle that asserts

$$\rho(x, A) \geq \rho(x, A \cup B)$$

for all menus $A, B \in \mathcal{M}$ and elements $x \in A$. It turns out to be equivalent for the function μ_A to satisfy

$$\mu_A(\theta) \geq \mu_{A \cup B}(\theta)$$

for all types $\theta \in \Theta$ that have the same maximal element in A and in $A \cup B$. This finding can be interpreted in terms of *self-selection*: types can increase their participation in the menu $A \cup B$, but only if they find better choices in the larger menu. For example, it should be reasonable for gamblers to show up in higher numbers to bookmakers that offer them more favorable odds on some sporting event. What's more, the expansion of bet types with the rise of online gambling can incentivize new customers to enter the market (Hing et al., 2022). Self-selection can also be observed in the dining industry. Garnett et al. (2019) found that

doubling the options of vegetarian meals increased sales of such meals by around 15%.

By contrast, self-selection cannot be detected in the general RUM examples because it can keep all Block-Marschak polynomials non-negative. In that example, the presence of self-selection makes the general RUM *misspecified* without rejecting this model outright. Our model does not have this identification issue and disentangles self-selection from the composition of the type space Θ .

More broadly, there are some behavioral patterns, such as reasoned-based heuristics in Shafir, Simonson, and Tversky (1993), where Monotonicity does not hold and hence, self-selection is no longer a plausible explanation. We use *context dependence* as a blanket term to refer to all such patterns (see a literature review by Rabin (1998)). An example of such patterns is extremeness aversion. Sharpe et al. (2008) observed that the removal of the largest drink size (44 oz) from the menu led consumers to shift their purchases from the 32-oz option—now the largest available size—to the 21-oz size.

Finally, we discuss applications where the type space Θ captures heterogeneous *risk attitudes* and *Cobb-Douglas* utility functions. For example, agents make more risk neutral choices when lotteries are presented in a binary format (i.e., only two states of the world have positive probabilities) instead of lotteries that are more complex or compound lotteries (Harrison et al., 2013). Changes in beliefs, which shall be discussed in a separate project, can arise in the form of partition dependence. Ahn and Ergin (2010) observed that agents, when choosing deductible plans, place a higher weight on surgeries when they are individually listed than when they are bundled under the umbrella term “surgery”. In such situations, the consumption domain X can be taken to consist of lotteries or Anscombe-Aumann (1963) uncertain prospects.

2 Preliminaries

Let $X = \{x, y, \dots\}$ be a non-singleton consumption domain. Assume that X is a convex subset of a normed linear space L .

For example, take any set Q of deterministic prizes (e.g. monetary ones). Let Δ^Q be the set of all *lotteries*: functions $x : Q \rightarrow [0, 1]$ such that $x(q) > 0$ is positive for finitely many $q \in Q$, and $\sum_{q \in Q} x(q) = 1$. If $X = \Delta^Q$, then X is a convex subset of the linear space $L = \mathbb{R}^Q$ endowed with the uniform norm. In the random expected utility model, GP take $X = \Delta^Q$ for some finite $Q = \{1, \dots, n\}$. In our framework, X need not be finite-dimensional, closed, or bounded, and elements in X may have interpretations other than lotteries.

A binary relation θ on X is called a *type* if it is complete, transitive, and satisfies *Inde-*

pendence: for all $x, y, z \in X$ and $\gamma \in (0, 1]$,

$$x\theta y \iff [\gamma x + (1 - \gamma)z]\theta[\gamma y + (1 - \gamma)z].$$

Let θ^* be the asymmetric part of the type θ : $x\theta^*y$ if $x\theta y$, but $y\theta x$ does not hold.¹

A type θ is called *null* if $x\theta y$ for all $x, y \in X$. Otherwise, θ is called *non-null*, and $x\theta^*y$ is strict for some $x, y \in X$.

Let $\mathcal{M} = \{A, B, \dots\}$ be the set of all *menus*—non-empty finite subsets of X . Singleton menus $\{x\}$ are written as x .

For any type θ and menu $A \in \mathcal{M}$, say that $x \in A$ is *rational* for θ in A if $x\theta y$ for all $y \in A$. Such x exists because θ is complete and transitive, and A is finite.

Let $\mathcal{T}$ be the set of all *total* types such that for all $x, y \in X$,

$$x\theta y\theta x \implies x = y.$$

This property implies that for any menu $A \in \mathcal{M}$ and total type $\theta \in \mathcal{T}$, there is a unique element $x_\theta(A) \in A$ that is rational for θ in A .

For any menu $A \in \mathcal{M}$ and element $x \in A$, let

$$\mathcal{T}_x(A) = \{\theta \in \mathcal{T} : x = x_\theta(A)\}$$

be the set of all types $\theta \in \mathcal{T}$ such that x is rational for θ in A . As $x_\theta(A)$ is unique for each $\theta \in \mathcal{T}$, then $\mathcal{T}$ is *partitioned* by the disjoint union

$$\mathcal{T} = \bigcup_{x \in A} \mathcal{T}_x(A). \quad (2)$$

For any $A, B \in \mathcal{M}$ and $\gamma \in [0, 1]$, define a mixture

$$\gamma A + (1 - \gamma)B = \{\gamma x + (1 - \gamma)y : x \in A, y \in B\}.$$

If $\gamma > 0$, then for any $x \in A$, the set $\mathcal{T}_x(A)$ is partitioned by

$$\mathcal{T}_x(A) = \bigcup_{y \in [\gamma x + (1 - \gamma)B]} \mathcal{T}_y(\gamma A + (1 - \gamma)B). \quad (3)$$

¹The more common notation $\succeq$ and $\succ$ for binary relations, such as types, is used occasionally in the proofs.

As $\gamma > 0$, then for any $\theta \in \mathcal{T}$, Independence implies

$$x_\theta(A) = x \iff x_\theta(\gamma A + (1 - \gamma)B) \in [\gamma x + (1 - \gamma)B],$$

and hence, the union (3) holds. It is disjoint because $x_\theta(\gamma A + (1 - \gamma)B)$ is determined uniquely by θ .

2.1 Mixture Linear Utility Maximization

A function $u : X \rightarrow \mathbb{R}$ is called *mixture linear* if for all $x, y \in X$ and $\gamma \in [0, 1]$,

$$u(\gamma x + (1 - \gamma)y) = \gamma u(x) + (1 - \gamma)u(y).$$

If $X = \Delta^Q$, then mixture linearity implies the *expected utility* form:

$$u(x) = \sum_{q \in Q} x(q)u(q) \text{ for all } x \in X.$$

In general, expected utility is not the only possible motivation for mixture linearity. For example, it holds for Cobb-Douglas utility functions over the positive orthant of consumption bundles in the logarithmic scale.

Let $\mathcal{U}$ be the set of all non-constant mixture linear functions $u : X \rightarrow \mathbb{R}$. For any $u \in \mathcal{U}$, let θ_u be the binary relation that is represented by u : for all $x, y \in X$,

$$x\theta_u y \iff u(x) \geq u(y).$$

A type θ is called *mixture continuous* if for all $x, y, z \in X$,

$$\{\gamma \in [0, 1] : z\theta[\gamma x + (1 - \gamma)y]\} \text{ and } \{\gamma \in [0, 1] : [\gamma x + (1 - \gamma)y]\theta z\}$$

are closed sets. Otherwise, θ is *discontinuous*.

Let $\mathcal{C}$ the class of all types that are mixture continuous and non-null. Herstein and Milnor (1953) *mixture space theorem* asserts that

$$\theta \in \mathcal{C} \iff \theta = \theta_u \text{ for some } u \in \mathcal{U}.$$

Moreover, such u is unique up to a positive linear transformation: if $\theta_u = \theta_v$ for some $u, v \in \mathcal{U}$, then $v = \alpha u + \beta$ for some $\alpha > 0$ and $\beta \in \mathbb{R}$.

However, total types can be mixture continuous only if X is a convex subset of a line

$\{x + \alpha d : \alpha \in \mathbb{R}\}$ for some $x, d \in L$ such that $d \neq 0$. Call such X *collinear*. Then the functions $u_+(x + \alpha d) = \alpha$ and $u_-(x + \alpha d) = -\alpha$ represent the only two non-null types θ_{u_+} and θ_{u_-} . Both are total and mixture continuous.

In the generic *non-collinear* case, X is not a subset of any line. Then all total types are discontinuous, that is, $\mathcal{T} \cap \mathcal{C} = \emptyset$. To show this, use non-collinearity to find $x, y, z \in X$ such that $x - y$ and $y - z$ are linearly independent. Let $\theta \in \mathcal{C}$. Then $\theta = \theta_u$ for some $u \in \mathcal{U}$. Without loss in generality, $u(x) \geq u(y) \geq u(z)$. Then either $u(x) = u(z)$, or $u(y) = u(\gamma x + (1 - \gamma)z)$ where $\gamma = \frac{u(y) - u(z)}{u(x) - u(z)}$. As $x \neq z$ and $y \neq \gamma x + (1 - \gamma)z$, then θ is not total in either case.

In a weaker form, the mixture space theorem can be extended to discontinuous (e.g. total) types. Say that $u \in \mathcal{U}$ is *maximized* by type θ if for all $x, y \in Z$,

$$x\theta y \implies u(x) \geq u(y). \quad (4)$$

Equivalently, $u \in \mathcal{U}$ is maximized by θ if for all $A \in \mathcal{M}$ and $x \in A$,

$$x \text{ is rational for } \theta \text{ in } A \implies x \text{ is a maximizer for } u \text{ in } A.$$

The one-way representation (4) with arbitrary utility functions is proposed by Richter and Peleg. Unlike their general case, mixture linearity allows to identify the function $u \in \mathcal{U}$ in (4) just as sharply as in the standard expected utility model.

A type θ is called *stable* if there exist $x^*, y^* \in X$ such that for any $z_x, z_y \in X$, there exists $\gamma \in (0, 1]$ such that

$$[\gamma z_x + (1 - \gamma)x^*]\theta^*[\gamma z_y + (1 - \gamma)y^*]. \quad (5)$$

The strict comparison $x^*\theta^*y^*$ is implied here because if $y^*\theta x^*$, then (5) fails for $z_x = x^*$ and $z_y = y^*$. Thus a stable type θ should allow some strict comparison $x^*\theta^*y^*$ that is unperturbed by mixtures $\gamma z_x + (1 - \gamma)x^*$ and $\gamma z_y + (1 - \gamma)y^*$, where the positive weight γ can be adapted to z_x and z_y . As γ can be arbitrarily small, then stability can be viewed as a local Archimedean property.

Theorem 1. *A type θ is stable if and only if θ maximizes some $u \in \mathcal{U}$. Such u is unique up to a positive linear transformation.*

Moreover, if X is finite dimensional, then all non-null types are stable.

All proofs are in the appendix. For any stable type θ , Theorem 1 asserts a mixture linear utility maximization (4) with the standard uniqueness claim for u . It also guarantees stability for all non-null types in finite dimensional frameworks. In general, a non-null type

θ can violate stability. Such θ can only have a trivial quasi-representation (4) where u is constant and hence, does not belong to $\mathcal{U}$.

The utility maximization (4) can be further refined via tie-breaking rules in Theorem 7 below. We also use that result to construct a rich family of stable total types on any convex domain X .

3 Main Representation Results

Let Ω be the set of all pairs (x, A) such that $A \in \mathcal{M}$ and $x \in A$, that is, x is a feasible element in a menu A . Such pairs are called *trials*.

A function $\rho : \Omega \rightarrow [0, 1]$ is called a *stochastic choice rule* (scr) if

$$\sum_{x \in A} \rho(x, A) = 1 \text{ for all } A \in \mathcal{M}. \quad (6)$$

Here, the probability $\rho(x, A)$ of any trial $(x, A) \in \Omega$ can be interpreted as the likelihood of x being chosen when the menu A is feasible.

A function $\mu : 2^{\mathcal{T}} \rightarrow \mathbb{R}_+$ is called a *discrete distribution* if

- (i) the set $\Theta = \{\theta \in \mathcal{T} : \mu(\theta) > 0\}$ is finite,
- (ii) for all sets $\Psi \subset \mathcal{T}$, $\mu(\Psi) = \sum_{\theta \in \Theta \cap \Psi} \mu(\theta)$,
- (iii) $\mu(\mathcal{T}) = \mu(\Theta) = 1$.

The set Θ is called the *support* of μ . Obviously, μ is a countably additive² probability measure, and its support Θ is determined uniquely.

Let $\mathcal{D}$ be the set of all discrete distributions. Say that a stochastic choice rule ρ is *represented* by a discrete distribution $\mu \in \mathcal{D}$ if for any trial $(x, A) \in \Omega$,

$$\rho(x, A) = \mu(\mathcal{T}_x(A)) \quad (7)$$

is the likelihood that μ assigns to all total types for which x is rational in A . By Theorem 1, stable total types $\theta \in \mathcal{T}$ with non-null closures can be interpreted in terms of expected utility maximization. Thus we refer to representation (7) as *discrete random expected utility* (DREU).

²For discrete distributions, countable and finite additivity are equivalent.

To get some observable implications of DREU, take any $(x, A) \in \Omega$, $B \in \mathcal{M}$, and $\gamma \in (0, 1]$. Let

$$\rho(x, A, \gamma, B) = \sum_{y \in [\gamma x + (1-\gamma)B]} \rho(y, \gamma A + (1-\gamma)B)$$

be the probability of all choices in the menu $\gamma A + (1-\gamma)B$ that belong to its submenu $\gamma x + (1-\gamma)B$.

Axiom 1 (Additivity). $\rho(x, A) = \rho(x, A, \gamma, B)$.

This equality follows from (7) that applies the discrete distribution μ to both sides of the partition (3). In particular, Additivity requires

Axiom 2 (Monotonicity). $\rho(x, A) \geq \rho(x, A \cup B)$.

Monotonicity holds in Block and Marschak's general random utility model and all of its refinements that take the distribution μ to be invariant across menus A . It holds in DREU as well, but need not be assumed separately in the presence of Additivity, which implies

$$\rho(x, A) = \rho(x, A, 0.5, A \cup B) \geq \rho(x, A \cup B, 0.5, A) = \rho(x, A \cup B).$$

Here the middle inequality holds because $0.5x + 0.5(A \cup B) \supset 0.5x + 0.5A$.

Next, DREU implies that for any $(x, A) \in \Omega$,

$$\rho(x, A) > 0 \iff x = x_\theta(A) \text{ for some } \theta \in \Theta, \tag{8}$$

where Θ is the finite support of μ . For any menu A , define the *capacity* of ρ in A

$$\kappa_\rho(A) = |\{x \in A : \rho(x, A) > 0\}|$$

as the number of all elements that are chosen with a positive probability in A . As Θ is finite, then by (8), $\kappa_\rho(A) \leq |\Theta|$.

Axiom 3 (Discreteness). *There is $n \in \mathbb{N}$ such that $\kappa_\rho(A) \leq n$ for all $A \in \mathcal{M}$.*

Discreteness cannot be rejected by finitely many observations and hence, plays a technical role akin to continuity conditions in GP.

Theorem 2. *A stochastic choice rule ρ satisfies Additivity and Discreteness if and only if ρ is represented by some discrete distribution $\mu \in \mathcal{D}$. Such μ is unique.*

The proof is constructive. By Discreteness, there is a maximal menu $B = \{b_1, \dots, b_k\}$ such that $\rho(b_i, B) > 0$ for all i . For any $x, y \in X$ and $i = 1, \dots, k$, let

$$x\theta_i y \iff \rho(\alpha x + (1 - \alpha)b_i, \alpha\{x, y\} + (1 - \alpha)B) > 0, \quad (9)$$

for sufficiently small $\alpha > 0$. Then Additivity delivers (7) where

$$\Theta = \{\theta_1, \dots, \theta_k\} \subset \mathcal{T} \quad \text{and} \quad \mu(\theta_i) = \rho(b_i, B) \text{ for all } i = 1, \dots, k.$$

All details are in the appendix.

Theorem 2 establishes that representation (7) is a *bijection* between $\mathcal{D}$ and the class of all stochastic choice rules that satisfy Additivity and Discreteness. Besides the existence and uniqueness claims, it remains to check that any discrete distribution $\mu \in \mathcal{D}$ represents some stochastic choice rule via (7). Indeed, the identity (6) for each menu $A \in \mathcal{M}$ follows from (7) by applying μ to both sides of the partition (2),

$$\sum_{x \in A} \rho(x, A) = \sum_{x \in A} \mu(\mathcal{T}_x(A)) = \mu(\mathcal{T}) = 1.$$

To illustrate the role of total types in Theorem 2, consider another definition. For any $\theta \in \mathcal{T}$ and $(x, A) \in \Omega$, let

$$\rho_\theta(x, A) = \begin{cases} 1 & \text{if } x = x_\theta(A) \\ 0 & \text{if } x \neq x_\theta(A). \end{cases}$$

Obviously, ρ_θ is a stochastic choice rule, and representation (7) asserts that

$$\rho = \sum_{\theta \in \Theta} \mu(\theta) \rho_\theta$$

is a convex combination of ρ_θ across $\theta \in \Theta$. By contrast, there is no convenient way to specify ρ_θ when $\theta \in \mathcal{C}$ because of tie-breaking. For example, the uniform tie-breaking rule violates both Discreteness and Additivity. Indeed, let X be non-collinear and $\theta \in \mathcal{C}$. Then there are distinct $x, y \in X$ such that $u(x) = u(y)$. Let $B = \{x, y\}$, $z = 0.5x + 0.5y$, and $C = 0.5B + 0.5B = \{x, y, z\}$. The uniform tie-breaking implies that

$$\begin{aligned} \rho_\theta(x, B) &= \rho_\theta(y, B) = \frac{1}{2}, \\ \rho_\theta(x, C) &= \rho_\theta(y, C) = \rho_\theta(z, C) = \frac{1}{3}, \end{aligned}$$

which violates Additivity because $\theta(x, B) \neq \theta(x, B, 0.5, B)$. Next, for any $n \in \mathbb{N}$, consider

a menu $A = \{a_1, \dots, a_n\}$ where $a_i = \frac{i}{n}x + \frac{n-1}{n}y$. The uniform tie-breaking implies that $\rho(a_i, A) = \frac{1}{n}$ for each i . As n is arbitrary, then Discreteness fails.

In Section 4, we compare DREU and Theorem 2 with GP's REU and their representation results.

3.1 Discrete Map Expected Utility

Additivity and representation (7) can be problematic if the distribution of types in Θ is context dependent. For example, adding extra options to a menu A can increase the probability of choosing $x \in A$ due to reason-based heuristics (e.g., Shafir, Simonson, and Tversky (1993)) and other behavioral reasons. Thus Monotonicity and a fortiori, Additivity can be violated.

To accommodate context dependence, discrete distributions can be assigned contingently on the feasible menu A via some function $\mu : \mathcal{M} \rightarrow \mathcal{D}$. It is convenient to write the distribution that μ assigns to A as μ_A rather than $\mu(A)$.

To impose continuity on μ , metrize $\mathcal{D}$ and $\mathcal{M}$ by the uniform and Hausdorff metrics respectively so that such that for all $\mu, \pi \in \mathcal{D}$ and $A, B \in \mathcal{M}$,

$$d_u(\mu, \pi) = \max_{\theta \in \mathcal{T}} |\mu(\theta) - \pi(\theta)|$$

$$d_h(A, B) = \max \left\{ \max_{x \in A} \min_{y \in B} \|x - y\|, \max_{y \in B} \min_{x \in A} \|x - y\| \right\}.$$

A continuous function $\mu : \mathcal{M} \rightarrow \mathcal{D}$ is called a *discrete map* if for all $A, B \in \mathcal{M}$, μ_A and μ_B have the same support

$$\Theta = \{\theta \in \mathcal{T} : \mu_A(\theta) > 0\} = \{\theta \in \mathcal{T} : \mu_B(\theta) > 0\}.$$

Let $\mathcal{D}(\mathcal{M})$ be the set of all discrete maps. With some abuse of notation, any discrete distribution $\mu \in \mathcal{D}$ is identified with the constant discrete map $\mu \in \mathcal{D}(\mathcal{M})$.

Say that a stochastic choice rule ρ is represented by a discrete map $\mu \in \mathcal{D}(\mathcal{M})$ if for all trials $(x, A) \in \Omega$,

$$\rho(x, A) = \mu_A(\mathcal{T}_x(A)) \tag{10}$$

is the likelihood that is assigned by μ_A to all total types θ for which x is rational in A . Obviously, (7) is a special case of (10) where the discrete map μ is constant.

Call the general representation (10) *discrete-map expected utility* (DMEU). This model violates Additivity because it applies distinct distributions μ_A and $\mu_{\gamma A + (1-\gamma)B}$ on the two sides of the partition (3). Similarly, Monotonicity can fail because the distributions μ_A and

$\mu_{A \cup B}$ need not be the same.

Instead, DMEU implies a weaker form of Additivity. Let $\{0 < \gamma_n \leq 1\}$ and $\{B_n \in \mathcal{M}\}$ denote sequences $\gamma_1, \gamma_2, \dots$ and $B_1, B_2, \dots$ such that $\gamma_n \in (0, 1]$ and $B_n \in \mathcal{M}$ for all $n \in \mathbb{N}$.

Axiom 4 (Slack Additivity (SA)). *For all trials $(x, A) \in \Omega$ and convergent sequences $\{0 < \gamma_n \leq 1\}$ and $\{B_n \in \mathcal{M}\}$*

$$\rho(x, A) > 0 \iff \lim_{n \rightarrow \infty} \rho(x, A, \gamma_n, B_n) > 0.$$

Given Additivity, SA is trivial because $\rho(x, A, \gamma_n, B_n) = \rho(x, A)$ for all n . More broadly, SA follows from DMEU, where

$$\lim_{n \rightarrow \infty} \rho(x, A, \gamma_n, B_n) = \lim_{n \rightarrow \infty} \mu_{\gamma_n A + (1-\gamma_n) B_n}(\mathcal{T}_x(A)) = \mu_{\gamma A + (1-\gamma) B}(\mathcal{T}_x(A))$$

because $\mu : \mathcal{M} \rightarrow \mathcal{D}$ is continuous, and

$$\mu_A(\mathcal{T}_x(A)) > 0 \iff \mu_{\gamma A + (1-\gamma) B}(\mathcal{T}_x(A)) > 0$$

because μ_A and $\mu_{\gamma A + (1-\gamma) B}$ share the same support $\Theta \subset \mathcal{T}$.

To get some non-asymptotic consequences from SA, suppose that the sequences $\gamma_n = \gamma \in (0, 1]$ and $B_n = B \in \mathcal{M}$ are constant for all $n \in \mathbb{N}$. Then SA implies the equivalence

$$\rho(x, A) > 0 \iff \rho(x, A, \gamma, B) > 0. \tag{11}$$

In DMEU, either of these two inequalities holds if and only if x is rational in A for some type $\theta \in \Theta$ in the support of the discrete map μ . Rewrite (11) as

$$\rho(x, A) = 0 \iff \rho(x, A, \gamma, B) = 0.$$

Thus SA still requires Additivity in the boundary case when either side in the equation $\rho(x, A) = \rho(x, A, \gamma, B)$ is zero.

Theorem 3. *A stochastic choice rule ρ satisfies SA and Discreteness if and only if ρ is represented by a discrete map $\mu \in \mathcal{D}(\mathcal{M})$. Such μ is unique, and its support Θ satisfies*

$$|\Theta| = \max_{A \in \mathcal{M}} \kappa_\rho(A).$$

This result characterizes DMEU by relaxing the Additivity axiom. The size $|\Theta|$ of the support Θ can be achieved as the maximal number of elements chosen with positive probability in some menu. A fortiori, the identity $|\Theta| = \max_{A \in \mathcal{M}} \kappa_\rho(A)$ holds in DREU.

Theorem 3 establishes that representation (10) is a *bijection* between $\mathcal{D}(\mathcal{M})$ and the class of all stochastic choice rules that satisfy SA and Discreteness. Similarly to DREU, any function ρ that is represented by some discrete map $\mu \in \mathcal{D}(\mathcal{M})$ via (10) must be a stochastic choice rule because for any menu $A \in \mathcal{M}$,

$$\sum_{x \in A} \rho(x, A) = \sum_{x \in A} \mu_A(\mathcal{T}_x(A)) = \mu_A(\mathcal{T}) = 1.$$

3.2 Self-Selection

Self-selection is a plausible form of context dependence for stochastic choices: types can increase their participation across distinct menus when their rational choices are improved. For example, it should be reasonable for gamblers to bet more actively with bookmakers who offer them more favorable odds.

Unlike the behavioral heuristics, like reason-based choice or extremeness aversion, self-selection should be consistent with Monotonicity. This condition appears particularly important for stochastic choice models because it is convenient for empirical tests and normative interpretations.

A discrete map $\mu \in \mathcal{D}(\mathcal{M})$ is called *selective* if for all menus $A, B \in \mathcal{M}$ and types $\theta \in \mathcal{T}$,

$$x_\theta(A) = x_\theta(A \cup B) \implies \mu_A(\theta) \geq \mu_{A \cup B}(\theta). \quad (12)$$

Condition (12) asserts that all types $\theta \in \Theta$ that preserve their rational choice $x_\theta(A)$ when A is increased to the larger menu $A \cup B$ should not become more likely in $A \cup B$. This condition is plausible under self-selection because such types θ should not be motivated to increase their participation in $A \cup B$, while no other type $\psi \in \Theta$ should be motivated to decrease its participation $A \cup B$ because $x_\psi(A)$ is still feasible when A is replaced with $A \cup B$.

Theorem 4. *A stochastic choice rule ρ satisfies Monotonicity, SA, and Discreteness if and only if ρ is represented by a selective discrete map μ .*

This result asserts that given Monotonicity, all representations by discrete maps can be interpreted in terms of self-selection. As a special case of DMEU, the uniqueness and non-redundancy claims from Theorem 3 apply as is in the self-selective case.

In general, the selective property (12) does not have to belong to RUM. Let $X = \{a, b, c, d\}$ and three types in Θ : $\succ_a = a \succ b \succ c \succ d$, $\succ_b = b \succ a \succ c \succ d$ and $\succ_d = d \succ a \succ b \succ c$. Consider the following four menus and distributions on types. The item in the parentheses denotes the maximizer.

menu A	$\mu_A(\succ_a)$	$\mu_A(\succ_b)$	$\mu_A(\succ_d)$
$\{a, b, c, d\}$	0.4 (a)	0.5 (b)	0.1 (d)
$\{a, c, d\}$	0.5 (a)	0.1 (a)	0.4 (d)
$\{b, c, d\}$	0.1 (b)	0.5 (b)	0.4 (d)
$\{c, d\}$	0.2 (c)	0.2 (c)	0.6 (d)

This dataset satisfies the selective property (12). The Block-Marschak polynomial $q(d, \{c, d\})$, which is interpreted as the probability that d is the best element in $\{c, d\}$ but worse in $\{a, b, d\}$ would be computing as this:

$$q(d, \{c, d\}) = \rho(d, \{c, d\}) - \rho(d, \{a, c, d\}) - \rho(d, \{b, c, d\}) + \rho(d, X) = 0.6 - 0.4 - 0.4 + 0.1 = -0.1.$$

According to the non-negativity condition by Falmagne (1978), this stochastic choice rule does not belong to RUM.

Consider now a separate case where Block-Marschak polynomials are all non-negative (and hence belongs to RUM): Let $X = \{a, b, c\}$ and Θ include two types: $a\theta b\theta c$ and $c\psi b\psi a$, each taking up half of the true population. Assume a type would participate for sure if their favorite option is available, and with probability 50% otherwise. In menu $A = \{a, b\}$, all type θ agents will show up whereas half of type ψ agents will show up. θ agents will choose a and ψ agents will choose b , and the market will observe a stochastic choice rule of $\rho(a, A) = \frac{2}{3}$ and $\rho(b, A) = \frac{1}{3}$.³

However, researchers only observe the stochastic choice rule, and are not privy to the true distribution of types in the world. The following distribution would also provide the same ρ .⁴

$$\hat{\mu}(\theta) = \frac{5}{9}, \quad \hat{\mu}(\psi) = \frac{1}{3}, \quad \hat{\mu}(\xi) = \frac{1}{9}$$

where $b\xi c\xi a$. This is due to the non-uniqueness of Block-Marschak polynomials (Fishburn, 1998). As a result, the distribution of types can be misspecified in a random utility model where there is no parametric form, i.e., where agents preferences are arbitrary orders on the consumption space.

$$\begin{aligned} {}^3\rho(a, A) &= \frac{\frac{1}{2}}{\frac{1}{2} + \frac{1}{2} \times \frac{1}{2}}, \rho(b, A) = \frac{\frac{1}{2} \times \frac{1}{2}}{\frac{1}{2} + \frac{1}{2} \times \frac{1}{2}} \\ {}^4\rho(a, A) &= \frac{\frac{5}{9}}{\frac{5}{9} + \frac{1}{3} \times \frac{1}{2} + \frac{1}{9}} = \frac{2}{3}, \rho(b, A) = \frac{\frac{1}{3} \times \frac{1}{2} + \frac{1}{9}}{\frac{5}{9} + \frac{1}{3} \times \frac{1}{2} + \frac{1}{9}} = \frac{1}{3}. \end{aligned}$$

4 Discussion

DREU can be naturally compared with GP's random expected utility. In their model, $X = \Delta^n$ is a finite-dimensional simplex, and for all $(x, A) \in \Omega$,

$$\rho(x, A) = \mu(\mathcal{C}_x(A)), \quad (13)$$

where μ is a finitely additive probability measure over $\mathcal{C}$, and

$$\mathcal{C}_x(A) = \{\theta \in \mathcal{C} : x \text{ is rational for } \theta \text{ in } A\}.$$

Note that $\mathcal{C} = \bigcup_{x \in A} \mathcal{C}_x(A)$, but this union need not be disjoint because a mixture continuous type $\theta \in \mathcal{C}$ can have several rational elements in A . To make representation (13) consistent with normalization (6), it must be assumed that the probability of ties is zero in each menu A , that is,

$$\mu(\mathcal{C}_x(A) \cap \mathcal{C}_y(A)) = 0 \quad (14)$$

for all distinct $x, y \in A$. GP calls such distributions μ *regular*. Regularity prohibits μ to have a finite support, except for the case when X is collinear and there are only two possible types.

By contrast, our representation (7) uses total rather than mixture continuous types, and takes μ to be discrete. This change in mathematical structure allows to identify μ more directly via the algebraic formula (9) rather than GP's analytical arguments. Our approach is more flexible about the consumption domain: X is convex, but need not be closed, or bounded, or finite dimensional. For example, X may consist of monetary gambles with payoffs on the real line, or X can be the non-negative orthant of consumption bundles.

GP's REU implies Additivity. Similarly to partition (3) for total types, one can derive from (13) that for all $(x, A) \in \Omega$, $B \in \mathcal{M}$, and $\gamma \in (0, 1]$.

$$\mathcal{C}_x(A) = \bigcup_{y \in [\gamma x + (1-\gamma)B]} \mathcal{C}_y(\gamma A + (1-\gamma)B),$$

but this union need not be disjoint because rational elements are not unique for mixture continuous types. Additivity still follows from representation (13) that applies μ to both sides of the above union and assumes the no-tie constraint (14).

On the other hand, our DREU implies all axioms in GP's Theorem 2. For any menu $A \in \mathcal{M}$, let ρ_A be the discrete distribution over X such that $\rho_A(x) = \rho(x, A)$ if $x \in A$ and $\rho_A(x) = 0$ if $x \notin A$. Endow the space of such distributions with the topology of weak

convergence. Besides Monotonicity, GP's Theorem 2 requires the following conditions for all $A, B \in \mathcal{M}$, $x, y \in X$, and $\gamma \in [0, 1]$,

- (i) Linearity: if $\gamma > 0$ and $x \in A$, then $\rho(x, A) = \rho(x, A, \gamma, y)$,
- (ii) Mixture Continuity: $\rho_{\gamma A + (1-\gamma)B}$ is continuous with respect to variations in γ ,
- (iii) Extremeness: if $x \neq y$ and $\gamma \in (0, 1)$, then

$$\rho(\gamma x + (1 - \gamma)y, \{x, y, \gamma x + (1 - \gamma)y\}) = 0.$$

Linearity is a special case of Additivity where $B = y$ is a singleton. Both MC and Extremeness are implied by DMEU and a fortiori, by DREU. Indeed, the mixture $\gamma x + (1 - \gamma)y$ where $\gamma \in (0, 1)$ cannot be rational for any total type θ in a menu A that contains both x and y . In the proof of Theorem 3, we derive MC from (10). Given Monotonicity, Extremeness implies that if z is chosen with a positive probability in a menu A , then z is an extreme point in A .

Therefore, if $X = \Delta^n$, and Discreteness holds, then the lists of axioms in our Theorem 2 and GP's Theorem 2 become equivalent. By contrast, their Theorem 3 requires that μ in (13) is countably additive, and ρ_A is continuous with respect to A . Such continuity is incompatible with DMEU because the stochastic choice rules ρ_θ are not continuous for total types θ , except when $X = \Delta^2$.

We conjecture that Additivity can be derived from GP's axioms even if X is not finite dimensional, but proving this claim appears complicated and tangential to our desiderata. So we skip this effort. Besides making the proofs and the statement of Theorem 2 more parsimonious, Additivity is conveniently relaxed to SA in Theorem 3 and its applications.

Note also that Additivity can be derived from its subadditivity part: for all $(x, A) \in \Omega$, $B \in \mathcal{M}$, and $\gamma \in (0, 1)$,

$$\rho(x, A) \geq \rho(x, A, \gamma, B).$$

Indeed, if any of these inequalities is strict, then by (6),

$$1 = \sum_{x \in A} \rho(x, A) > \sum_{x \in A} \sum_{y \in B} \rho(\gamma x + (1 - \gamma)y, \gamma A + (1 - \gamma)B) \geq 1.$$

By contradiction, Additivity holds.

4.1 Random Risk Attitudes

Olschewski et al. (2022) discovered that the random utility model does not precisely differentiate between risk-averse and risk-seeking agents. This motivates us to next extend the

discussion of menu dependence to risk attitudes.

Let $X = \Delta^Q$ be the set of all lotteries on some interval $Q \subset \mathbb{R}$ of monetary prizes.⁵ This interval is convex, but need not be closed or bounded. Any $q \in Q$ is associated with the degenerate lottery that delivers q with certainty. Accordingly, $Q \subset X$ denotes the subclass of such degenerate lotteries.

Interpret types $\theta \in \mathcal{T}$ in terms of risk attitudes. For each $x \in \Delta^Q$, let

$$E(x) = \sum_{q \in Q} x(q)q$$

be the mathematical expectation (i.e. the mean) of the lottery x .

Axiom 5 (Stable Risk Aversion (SRA)). *For any $q \in Q$ and $x \in \Delta^Q$,*

$$q > E(x) \implies \rho(x, \{x, q\}) = 0.$$

For any $q, r \in Q$ such that $q > r$, there is $\varepsilon > 0$ such that

$$\rho(r, \{r, \varepsilon r + (1 - \varepsilon)q\}) = 0.$$

A type $\theta \in \mathcal{T}$ is called *risk averse* if it maximizes some $u \in \mathcal{U}$ with a strictly increasing and concave restriction to Q . Let $\mathcal{T}_{RA}$ be the set of all risk averse types.

Corollary 5. *ρ satisfies SA, Discreteness, and SRA if and only if ρ is represented by a discrete map $\mu \in \mathcal{D}(\mathcal{M})$ such that its support $\Theta \subset \mathcal{T}_{RA}$ consists of risk averse types.*

By Theorem 3, the discrete map μ is unique. If Monotonicity or Additivity are imposed, then μ is selective or respectively, invariant of the menu A .

Peng (2025) explores the effects of a certain option (degenerate lottery) on risk behavior in an experimental setting. Let there be two types: a risk averse type θ with a Bernoulli utility index of $u_\theta(x) = \sqrt{x}$ and a risk neutral type n that always chooses the option with a higher expected value. Assume the following observations on two menus:

$$A = \left\{ r \equiv \begin{pmatrix} \$10 & \text{w.p.} & 0.5 \\ \$0 & \text{w.p.} & 0.5 \end{pmatrix}, l \equiv \begin{pmatrix} \$8 & \text{w.p.} & 0.5 \\ \$1 & \text{w.p.} & 0.5 \end{pmatrix} \right\} \text{ and } A' = A \cup \{c \equiv (\$3 \text{ w.p. } 1)\}.$$

A risk averse type will choose l in both cases whereas a risk neutral type will chose r . Let the data show that $\rho(r, A) = \rho(l, A) = 0.5$ but $\rho(r, A') = 0.25$ and $\rho(l, A') = 0.75$. This

⁵This example can be extended to the set X of all Borel distributions on Q with bounded rather than finite support. For brevity, we adopt the simplest framework here that still allows to identify a random risk aversion model.

would suggest that in menu A , the population consists of half risk neutral agents and half risk averse agents. However, in menu A' , risk neutral agents only takes up a quarter of the population while the rest are risk averse. This is a violation of Monotonicity as l is chosen with a higher likelihood in a superset menu. Hence, GP's standard REU would fail to model this phenomenon, and menu-dependence is needed.

We can explain this phenomenon by treating the certain option as a decoy, where its presence will make some agents more risk averse. Another working paper with a similar design (Lim and R., 2021) confirms that adding a safer option increases risk aversion. They also observed that adding a riskier option does not have a noticeable change in risk attitude. The latter was also noticed by Chen et al. (2024).

This can be extended to more types. Specifically, we can use discrete Arrow-Pratt indices by Baillon and L'Haridon (2021) to capture the finite nature of the type space. A discrete lognormal distribution may be appropriate for $\mu_A(\cdot)$ as CRRA, in the format of $u(x) = \frac{x^r}{1-r}$, indices are positive and skewed right, with most agents exhibiting some level of risk aversion. The mode of μ_A will move left (towards 0) when a certain option is present.

More broadly, X can be a set $\Delta^{Q \times S}$ of all *acts*—functions $x : S \rightarrow \Delta^Q$ on a finite state space S . This domain is introduced by Anscombe and Aumann (1963) (henceforth, AA), and can be embedded as a convex compact subset of $L = \mathbb{R}^{Q \times S}$.

When X is the AA domain, then the heterogeneity of risk attitudes can be combined with heterogeneity of beliefs. The presence of random beliefs allows to model learning patterns that can be affected by the composition of the menu A . We delegate the discussion of such learning to a companion paper.

4.2 Random Cobb-Douglas

Representations from Theorems 2–4 can be adapted to characterize random Cobb-Douglas utility in the setting where $X = \mathbb{R}_{++}^n$ is the positive orthant of consumption bundles in the logarithmic scale.

For any $x, y \in X$, write $x \gg y$ if $x_i > y_i$ for all $i = 1, \dots, n$.

Axiom 6 (Consumption Monotonicity (CM)). *For any $x, y \in \mathbb{R}_{++}^n$,*

$$y \gg x \implies \rho(x, \{x, y\}) = 0.$$

A type $\theta \in \mathcal{T}$ is called *Cobb-Douglas* if it maximizes some $u \in \mathcal{U}$ such that for all $x \in \mathbb{R}_{++}^n$,

$$u(x) = \sum_{i=1}^n \alpha_i x_i$$

for some $\alpha_i \geq 0$. Let $\mathcal{T}_{CD}$ be the set of all Cobb-Douglas types.

Corollary 6. ρ satisfies SA, Discreteness, and CM if and only if ρ is represented by a discrete map $\mu \in \mathcal{D}(\mathcal{M})$ such that its support $\Theta \subset \mathcal{T}_{CD}$ consists of Cobb-Douglas types.

The uniqueness of μ is again inherited from Theorem 3. The selective and invariant structures can be derived from Monotonicity and Additivity axioms respectively.

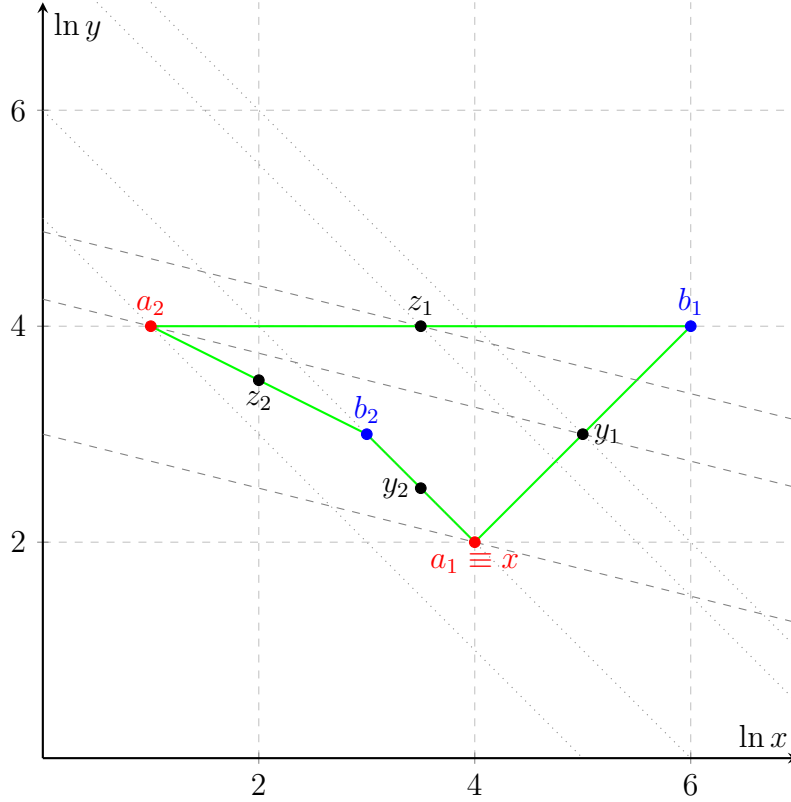


Figure 1: Illustration of Additivity with Cobb-Douglas types.

In Figure 1, two menus are $A = \{a_1 \equiv x = (4, 2), a_2 = (1, 4)\}$ and $B = \{b_1 = (6, 4), b_2 = (3, 3)\}$. Let a mixture be constructed with $\gamma = 0.5$. By additivity, the proportion of agents that choose $a_1 \equiv x$ among the red dots should be equal to the proportion of agents that choose either $y_1 \equiv \gamma x + (1 - \gamma)b_1$ or $y_2 \equiv \gamma x + (1 - \gamma)b_2$ among the black dots.

The essence of Cobb-Douglas utility functions are indifference curves that are parallel, provided commodities are in logs. A maximizer will choose the bundle on the indifference curve furthest from the origin. Different types are represented by different slopes. Consider two types respectively with slopes $m_1 = -1$ (illustrated by dotted lines) and $m_2 = -\frac{1}{4}$ (illustrated by dashed lines). All type 1 agents will choose $a_1 \equiv x$ in menu A while the type 2 agents will choose a_2 . In the mixture menu, type 1 agents will choose y_1 and type 2 will choose z_1 . This provides equal proportions in $\rho(x, A)$ and $\rho(x, A, \gamma, B)$.

4.3 Tie-Breakers

Next, we clarify the structure of stable total types. For any $u \in \mathcal{U}$, let

$$L_u = \{\alpha(x - y) : \alpha \geq 0 \text{ and } x, y \in X \text{ such that } u(x) = u(y)\}.$$

As X is convex, then L_u is a linear space.⁶

Theorem 7. *A type θ is stable if and only if there is a function $u \in \mathcal{U}$ and a type ψ in L_u such that for all $x, y \in X$,*

$$x\theta y \iff \begin{aligned} &u(x) > u(y), \text{ or} \\ &u(x) = u(y) \text{ and } (x - y)\psi 0. \end{aligned} \quad (15)$$

If (15) holds with another pair (u', ψ') , then $u' = \alpha u + \beta$ for some $\alpha > 0$ and $\beta \in \mathbb{R}$, $L_u = L_{u'}$, and $\psi = \psi'$.

The proof is in the appendix. Representation (15) asserts that any stable type θ maximizes some utility function $u \in \mathcal{U}$, which is unique up a positive linear transformation, and then uses the unique residual type ψ on the linear subspace L_u to compare any $x, y \in X$ such that $u(x) = u(y)$.

Representation (15) in Theorem 7 implies that for any stable type θ ,

$$\theta \text{ is total in } X \iff \psi \text{ is total in } L_u.$$

Indeed, if $a \sim_\psi b$ for some distinct elements $a, b \in L_u$, then by Independence, $\frac{a-b}{2} \sim_\psi 0$. Take $x, y \in X$ and $\alpha \geq 0$ such that $\alpha(x - y) = \frac{a-b}{2} \neq 0$ and $u(x) = u(y)$. By Independence, $x - y \sim_\psi 0$. By (15), $x \sim_\theta y$, and θ is not total.

For any $u \in \mathcal{U}$, let $\mathcal{T}_u$ be the set of all total types on L_u . Such types are called *tie-breakers*. To construct tie-breakers, take any *basis* H in the linear space L_u . By definition, $H \subset L_u$ is a linearly independent system of vectors that spans L_u . This definition is standard if L_u is finite dimensional. More generally, such H is known as a *Hamel basis* when L_u is infinite dimensional. The existence of a Hamel basis in any linear space is guaranteed by Zorn's lemma (see Aliprantis and Border (1999, Theorem 1.6)). For any $x \in L_u$ and $h \in H$, let

$$\alpha(x - y) + \beta(x' - y') = (\alpha + \beta) \left[\left(\frac{\alpha}{\alpha + \beta} x + \frac{\beta}{\alpha + \beta} x' \right) - \left(\frac{\alpha}{\alpha + \beta} y + \frac{\beta}{\alpha + \beta} y' \right) \right]$$

also belongs to L_u because $u \left(\frac{\alpha}{\alpha + \beta} x + \frac{\beta}{\alpha + \beta} x' \right) = u \left(\frac{\alpha}{\alpha + \beta} y + \frac{\beta}{\alpha + \beta} y' \right)$.

⁶Indeed, take any $\alpha \geq 0$ and $x, y \in X$ such that $u(x) = u(y)$. Then $\gamma\alpha(x - y) \in L_u$ for all $\gamma \in \mathbb{R}$ because if $\gamma < 0$, then $\gamma\alpha(x - y) = (-\gamma\alpha)(y - x)$. For any $\beta \geq 0$ and $x', y' \in X$ such that $u(x') = u(y')$, the sum

$x(h)$ be the coefficient (coordinate) of the vector h in the unique linear combination of basis vectors that equals x . Endow H with any total order $\geq$. Define a binary relation ψ_H for all $x, y \in L_u$ via

$$x\psi_H y \iff x = y \text{ or } x(h(x, y)) \geq y(h(x, y)), \quad (16)$$

where $h(x, y) \in H$ is the first vector in H such that $x(h) \neq y(h)$. Then ψ_H compares the coordinates of the vectors x and y *lexicographically* in the basis H . By definition, ψ_H is a tie-breaker: it is complete, total, transitive, and satisfies Independence.

Thus a rich family of total types can be generated by representations (15) and (16). These representations can be further refined (e.g. Fishburn (1982), Myerson (1986), Blume, Branderburger, and Dekel (1991)) by imposing more structure on the consumption domain X , the functions $u \in \mathcal{U}$, and the basis H .

4.4 Non-Redundant Type Spaces

Next, Discreteness can be strengthened to interpret the support Θ of the discrete map μ in terms of distinct expected utility functions.

Take any non-singleton menu A . Say that $x \in A$ is a *stable choice* in A if for any $y \in A \setminus x$ and $z_x, z_y \in X$, there is $\varepsilon > 0$ such that for all $\gamma \in [0, \varepsilon)$

$$\rho(\gamma z_x + (1 - \gamma)x, \{\gamma z_x + (1 - \gamma)x, \gamma z_y + (1 - \gamma)y\} \cup (A \setminus x)) > 0$$

Let $s_\rho(A)$ be the number of stable choices in A . If $\gamma = 0$, then the definition of a stable choice implies $\rho(x, A) > 0$. Thus $s_\rho(A) \leq k_\rho(A)$.

Axiom 7 (Stable Discreteness (SD)). *There is $A \in \mathcal{M}$ such that $s_\rho(A) \geq k_\rho(B)$ for all $B \in \mathcal{M}$.*

Clearly, SD implies Discreteness for $n = s_\rho(A)$. SD requires that the maximal number of chosen elements in an arbitrary menu B can be achieved via stable choices in some A .

A finite collection of total types $\Theta \subset \mathcal{T}$ is called *non-redundant* if all types $\theta_1, \theta_2 \in \Theta$ are stable and do not maximize the same mixture linear utility $u \in \mathcal{U}$. In other words, non-redundancy requires that distinct types should maximize distinct mixture linear utility functions $u_1, u_2 \in \mathcal{U}$ rather than differ only in tie-breaking rules.

Theorem 8. *If ρ is represented by a discrete map $\mu \in \mathcal{M}(\mathcal{D})$, then the support Θ is non-redundant if and only if ρ satisfies SD.*

A APPENDIX: PROOFS

Theorem 3 is shown first because it is the main step in the proofs of our representation results for stochastic choice functions ρ .

Let ρ satisfy SA and Discreteness. For all $A \in \mathcal{M}$, let

$$\Phi(A) = \{x \in A : \rho(x, A) > 0\}$$

be the set of all elements in A that are chosen with a positive probability.

For all $(x, A) \in \Omega$, $B \in \mathcal{M}$, $y \in X$, and $\gamma \in (0, 1]$, SA implies

(SA1) $\rho(x, A) > 0$ if and only if $\rho(x, A, \gamma, B) > 0$,

(SA2) $\rho(x, A) = 0$ if and only if $\rho(x, A, \gamma, B) = 0$,

(SA3) if $x \in \Phi(A \cup B)$, then $x \in \Phi(A)$,

(SA4) $\Phi(\gamma A + (1 - \gamma)y) = \gamma\Phi(A) + (1 - \gamma)y$.

By SA, $\rho(x, A) > 0$ implies that

As $0.5x + 0.5A \subset 0.5x + 0.5(A \cup B)$, then by (??)

$$\rho(x, A \cup B, 0.5, A) \leq \rho(x, A, 0.5, A \cup B).$$

By SA, $\rho(x, A \cup B) > 0$ implies $\rho(x, A, 0.5, A \cup B) > 0$ and hence,

$$\begin{aligned} \rho(x, A \cup B) > 0 &\implies \rho(x, A \cup B, 0.5, A) > 0 \implies \\ \rho(x, A, 0.5, A \cup B) > 0 &\implies \rho(x, A) > 0. \end{aligned} \tag{17}$$

If $B = y$ is a singleton for some $y \in X$, then SA1 implies that

$$\rho(x, A) > 0 \iff \rho(\gamma x + (1 - \gamma)y, \gamma A + (1 - \gamma)y) > 0.$$

It follows that

By SA, Φ satisfies equivalences

$$x \in \Phi(A) \iff [\gamma x + (1 - \gamma)B] \cap \Phi(\gamma A + (1 - \gamma)B) \neq \emptyset$$

Show that Φ satisfies the Chernoff condition (also known as Sen's α property): for all $A, B \in \mathcal{M}$ and $x \in A$,

By SA,

By Discreteness, there is a menu $B \in \mathcal{M}$ where ρ achieves its maximal capacity across all menus,

$$\kappa_\rho(B) = \max_{A \in \mathcal{M}} \kappa_\rho(A).$$

Let $C = \Phi(B)$ and $\kappa = |C|$. By Chernoff, $\Phi(C) = C$.

Lemma A.1. *There is a finite set $\Theta \subset \mathcal{T}$ such that for all $A \in \mathcal{M}$,*

$$\Phi(A) = \bigcup_{\theta \in \Theta} x_\theta(A).$$

Such Θ is unique.

To show this claim, assume first that X is bounded, and there is $R > 0$ such that $\|x\| < R$ for all $x \in X$. Let $R = \max_{i=1, \dots, k} \|x_i\|$. Take $\alpha \in (0, 1)$ such that

$$2\alpha R < (1 - \alpha) \min_{1 \leq i < j \leq k} \|x_i - x_j\|.$$

Then for any $\alpha \in (0, \alpha_{xy})$, the menu $\alpha\{x, y\} + (1 - \alpha)B$ has $2k$ distinct elements. To show this, it is enough to check that for any distinct $i, j \in \{1, \dots, k\}$, the menu

$$\alpha\{x, y\} + (1 - \alpha)\{x_i, x_j\}$$

has four distinct elements. Here the inequalities

$$\alpha x + (1 - \alpha)x_i \neq \alpha y + (1 - \alpha)x_i \neq \alpha y + (1 - \alpha)x_j \neq \alpha x + (1 - \alpha)x_j \neq \alpha x + (1 - \alpha)x_i$$

are obvious, the other two inequalities $\alpha x + (1 - \alpha)x_i \neq \alpha y + (1 - \alpha)x_j$ and $\alpha x + (1 - \alpha)x_j \neq \alpha y + (1 - \alpha)x_i$ follow from

$$(1 - \alpha)\|x_i - x_j\| - \alpha\|x - y\| > 0.$$

As $x_i \in \Phi(B)$ for all i , then by Type Independence, $\Phi(\alpha\{x, y\} + (1 - \alpha)B)$ overlaps with each of the k distinct sets

$$A_i = \alpha\{x, y\} + (1 - \alpha)x_i.$$

By (??), each of these k overlaps contains precisely one element, $\alpha x + (1 - \alpha)x_i$ or $\alpha y + (1 - \alpha)x_i$.

Lemma A.2. *For any finite set $\Theta = \{\theta_1, \dots, \theta_k\} \subset \mathcal{T}$, there is a menu $B = \{b_1, \dots, b_k\}$ such that for all $i = 1, \dots, k$,*

$$b_i = R_{\theta_i}(B).$$

Proof. Let I be the set of all pairs $i, j \in \{1, \dots, k\}$ such that $i > j$. For each pair $(i, j) \in I$, the types $\theta_i, \theta_j \in \mathcal{D}$ are distinct and total. Thus there are distinct elements $x_{ij}, x_{ji} \in X$ such that $x_{ij}\theta_i x_j$ and $x_{ji}\theta_j x_i$. For any $n \in \{1, \dots, k\}$, let

$$b_n = \sum_{(i,j) \in I} \frac{2}{k(k-1)} x_{ij}^n,$$

where $x_{ij}^n = x_{ij}$ if $x_{ij}\theta_n x_{ji}$ and $x_{ij}^n = x_{ji}$ if $x_{ji}\theta_n x_{ij}$. For any $(m, n) \in I$, the types θ_n, θ_m satisfy Independence and hence,

$$\begin{aligned} b_m &= \left(\sum_{(i,j) \in I} \frac{2}{k(k-1)} x_{ij}^m \right) \theta_m \left(\sum_{(i,j) \in I} \frac{2}{k(k-1)} x_{ij}^n \right) = b_n \\ b_n &= \left(\sum_{(i,j) \in I} \frac{2}{k(k-1)} x_{ij}^n \right) \theta_n \left(\sum_{(i,j) \in I} \frac{2}{k(k-1)} x_{ij}^m \right) = b_m \end{aligned} \tag{18}$$

because $x_{ij}^n \theta_n x_{ij}^m$ and $x_{ij}^m \theta_m x_{ij}^n$ for all $(i, j) \in I$. Moreover, when $i = m$ and $j = n$, then these comparisons are strict, and hence b_m, b_n are distinct elements.

Let $B = \{b_1, \dots, b_k\}$. By (18), $b_i \theta_i b_j$ for all i, j and hence, $b_i = R_{\theta_i}(B)$. $\square$

Take any stochastic choice rule ρ . For all menus $A \in \mathcal{M}$, let

$$R_\rho(A) = \{x \in A : \rho(x, A) > 0\}.$$

Suppose that ρ is represented by a discrete map μ . Let $\Theta \subset \mathcal{T}$ be the finite support of μ . By definition, for any $A \in \mathcal{M}$,

$$\Theta = \{\theta \in \mathcal{T} : \mu_A(\theta) > 0\} = \{\theta_1, \dots, \theta_k\}.$$

Representation (10) implies that for all $A \in \mathcal{M}$,

$$R_\rho(A) = \bigcup_{\theta \in \Theta} x_\theta(A) = \{x \in A : \Theta \cap \mathcal{T}_x(A) \neq \emptyset\} \tag{19}$$

consists of all elements that are rational in A for at least one type $\theta \in \Theta$. Representation refines the *multiple utility* model proposed by Aizerman and Malishevski Aizerman and Malishevski (1981) by restricting the type space Θ to $\mathcal{T}$ rather than to arbitrary total orders.

Check SA. By (10) and (3), for all $(x, A) \in \Omega$, $B \in \mathcal{M}$ and $\gamma \in (0, 1)$,

$$\begin{aligned} \rho(x, A) = 0 &\iff \Theta \cap \mathcal{T}_x(A) = \emptyset \implies \\ \Theta \cap \left[\bigcup_{y \in \gamma x + (1-\gamma)B} \mathcal{T}_y(\gamma A + (1-\gamma)B) \right] &= \emptyset \iff \\ \rho(x, A, \gamma, B) = \sum_{y \in \gamma x + (1-\gamma)B} \rho(y, \gamma A + (1-\gamma)B) &= 0. \end{aligned}$$

Check AA. Take any $(x, A) \in \Omega$ and converging sequences $\{\gamma_m\}_{m=1}^\infty \in (0, 1]$ and $\{B_n\}_{n=1}^\infty \in \mathcal{M}$. Let $\gamma = \lim_{m \rightarrow \infty} \gamma_m$ and $B = \lim_{n \rightarrow \infty} B_n$. By (10),

$$\rho(x, A) > 0 \implies \mu(\mathcal{T}_x(A)) > 0 \implies \mu_{\gamma A + (1-\gamma)B}(\mathcal{T}_x(A)) > 0.$$

The Hausdorff metric implies that

$$\lim_{m, n \rightarrow \infty} d_h(\gamma_m A + (1-\gamma_m)B_n, \gamma A + (1-\gamma)B) = 0$$

As μ is continuous, then

$$\lim_{m, n \rightarrow \infty} \rho(x, A, \gamma_m, B_n) = \lim_{m, n \rightarrow \infty} \mu_{\gamma_m A + (1-\gamma_m)B_n}(\mathcal{T}_x(A)) = \mu_{\gamma A + (1-\gamma)B}(\mathcal{T}_x(A)) > 0.$$

Discreteness holds because by (19), $|R_\rho(A)| \leq |\Theta|$. Moreover, by Lemma A.2, there is $B \in \mathcal{M}$ such that $|B| = |\Theta|$ and for each $b \in B$, there is $\theta \in \Theta$ such that $x_\theta(B) = b$. Then

$$|\Theta| = |R_\rho(B)| = \max_{A \in \mathcal{M}} |R_\rho(A)|.$$

Suppose that ρ is represented by another discrete map $\mu^* \in \mathcal{D}(\mathcal{M})$.

A.1 Other proofs for stochastic choice functions

A.2 Proof of Theorem 1

Take any type θ . Let θ maximize $u \in \mathcal{U}$. Then for all $x, y \in X$,

$$u(x) > u(y) \implies x \succ y. \tag{20}$$

As u is non-constant, then $u(x) > u(y)$ for some $x, y \in X$. For any $z_x, z_y \in X$, take $\gamma \in (0, 1]$ such that

$$\gamma u(z_x) + (1 - \gamma)u(x) > \gamma u(z_y) + (1 - \gamma)u(y).$$

By (20), $\gamma z_x + (1 - \gamma)x \succ \gamma z_y + (1 - \gamma)y$. Thus θ is stable.

Suppose instead that θ is stable and show that θ maximizes some $u \in \mathcal{U}$. Take any menu $A = \{a_1, a_2, \dots, a_n\}$ such that

$$a_1 \succeq a_2 \succeq \dots \succeq a_n.$$

Let X_A be the convex hull of A in L . As X is convex, then $X_A \subset X$. Show that for all $z \in X_A$,

$$a_1 \succeq z \succeq a_n. \tag{21}$$

The proof is by induction with respect to $n \geq 2$. If $n = 2$, then by Independence, (21) holds for all $z \in X_A$ because $z = \gamma a_1 + (1 - \gamma)a_2$ for some $\gamma \in [0, 1]$. Let $n > 2$. Take any $z \in X_A$ and write it as $z = \gamma y + (1 - \gamma)a_n$ for some $\gamma \in [0, 1]$ and $y \in X_{A \setminus a_n}$, where $X_{A \setminus a_n}$ is the convex hull of $A \setminus a_n$. By the inductive assumption, $a_1 \succeq y \succeq a_{n-1} \succeq a_n$. By Independence,

$$a_1 \succeq \gamma y + (1 - \gamma)a_1 \succeq \gamma y + (1 - \gamma)a_n \succeq \gamma a_n + (1 - \gamma)a_n = a_n$$

and hence, (21) holds.

Let $\mathcal{U}_A$ be the set of all non-constant mixture linear functions $u : X_A \rightarrow \mathbb{R}$.

Lemma A.3. *If $a_1 \succ a_n$, then there is $u \in \mathcal{U}_A$ such that for all $x, y \in X_A$,*

$$x \succeq y \implies u(x) \geq u(y). \tag{22}$$

Moreover, such u is unique up to a positive linear transformation.

Proof. Suppose that $a_1 \succ a_n$. For each $\alpha \in [0, 1]$, let $z_\alpha = \alpha a_1 + (1 - \alpha)a_n$. Show that for all $\alpha, \beta \in [0, 1]$,

$$\alpha \geq \beta \iff z_\alpha \succeq z_\beta. \tag{23}$$

As $a_1 \succ a_n$, then by Independence,

$$\begin{aligned} \alpha > \beta &\implies z_\alpha = \alpha a_1 + (1 - \alpha)a_n \succ a_n \implies \\ & z_\alpha = \frac{\beta}{\alpha} z_\alpha + \frac{\alpha - \beta}{\alpha} z_\alpha \succ \frac{\beta}{\alpha} z_\alpha + \frac{\alpha - \beta}{\alpha} a_n = z_\beta. \end{aligned}$$

If $\alpha \geq \beta$, then either $z_\alpha = z_\beta$ or $z_\alpha \succ z_\beta$. Thus (23) holds.

For all $x \in X_A$, let

$$\begin{aligned} u(x) &= \inf\{\alpha \in [0, 1] : z_\alpha \succeq x\} \\ v(x) &= \sup\{\beta \in [0, 1] : x \succeq z_\beta\}. \end{aligned} \tag{24}$$

By (21), these values are well-defined. Show that $u = v$. Indeed, $z_\alpha \succeq x \succeq z_\beta$ implies $\alpha \geq \beta$ and hence, $u(x) \geq v(x)$. Suppose that $u(x) > v(x)$. Then there is γ such that $u(x) > \gamma > v(x)$. By (24), $u(x) > \gamma$ implies $x \succ z_\gamma$. By (24), $\gamma > v(x)$ implies $z_\gamma \succ x$. By contradiction, $u(x) = v(x)$.

By (24), u is not constant because $u(a_1) = 1$ and $u(a_n) = 0$. Show that u is mixture linear. By Independence, for any $x, y \in X$ and $\alpha, \beta, \gamma \in [0, 1]$ such that $z_\alpha \succeq x$ and $z_\beta \succeq y$,

$$z_{\gamma\alpha + (1-\gamma)\beta} = \gamma z_\alpha + (1-\gamma)z_\beta \succeq \gamma x + (1-\gamma)z_\beta \succeq \gamma x + (1-\gamma)y.$$

Thus $u(\gamma x + (1-\gamma)y) \leq \gamma\alpha + (1-\gamma)\beta$. By definition of u , α and β can be arbitrarily close to $u(x)$ and $u(y)$ respectively. Thus

$$u(\gamma x + (1-\gamma)y) \leq \gamma u(x) + (1-\gamma)u(y).$$

Analogously, $v(\gamma x + (1-\gamma)y) \geq \gamma v(x) + (1-\gamma)v(y)$. As $u = v$, then $u \in \mathcal{U}_A$.

For all $x, y \in X_A$, let $\gamma = 0.5u(x) + 0.5u(y)$. By (24),

$$u(x) > u(y) \implies u(x) > \gamma > v(y) \implies x \succ z_\gamma \succ y \implies x \succ y.$$

Thus representation (22) holds.

Suppose that (22) holds with another $w \in \mathcal{U}_A$. As $a_1 \succeq a_n$, then $w(a_1) \geq w(a_n)$. If $w(a_1) = w(a_n)$, then by (21), $w(a_1) \geq w(z) \geq w(a_n)$ for all $z \in X_A$, and w is constant on X_A . By contradiction, $w(a_1) > w(a_n)$. For all $x \in X_A$,

$$w'(x) = \frac{w(x) - w(a_n)}{w(a_1) - w(a_n)}.$$

Then $w' \in \mathcal{U}_A$ is a positive linear transformation of w such that $w'(a_1) = 1$ and $w'(a_n) = 0$. Representation (22) still holds for w' . As $\alpha = w'(z_\alpha) \geq w'(x)$ for all α such that $z_\alpha \succeq x$, then $u(x) \geq w'(x)$. As $\beta = w'(z_\beta) \leq w'(x)$ for all β such that $x \succeq z_\beta$, then $w'(x) \geq v(x)$. Thus $w'(x) = u(x)$, and w is a positive linear transformation of u . $\square$

As $\succeq$ is stable, then there exist $x^*, y^* \in X$ that satisfy (5). Take any menu $A =$

$\{a_1, a_2, \dots, a_n\}$ such that $x^*, y^* \in A$, and

$$a_1 \succeq a_2 \succeq \dots \succeq a_n.$$

As $x^* \succ y^*$, then $a_1 \succ a_n$. By Lemma A.3, there is $u \in \mathcal{U}_A$ such that (22) holds. By (22), $u(x^*) \geq u(y^*)$. Suppose that $u(x^*) = u(y^*)$. Then for all $\gamma \in (0, 1]$,

$$u(\gamma y^* + (1 - \gamma)a_1) > u(\gamma x^* + (1 - \gamma)a_n)$$

and hence, $\gamma y^* + (1 - \gamma)a_1 \succ \gamma x^* + (1 - \gamma)a_n$. Thus $y^* \succeq^* x^*$. This contradicts (5). Thus, $u(x^*) > u(y^*)$.

By Lemma A.3, there is a unique $u_A \in \mathcal{U}_A$ that satisfies (22), and

$$u_A(x^*) = 1 > 0 = u_A(y^*).$$

For each $x \in X$, define

$$u(x) = u_{\{x^*, y^*, x\}}(x).$$

Take any $x, y \in X$ and $\gamma \in [0, 1]$. Let $B = \{x^*, y^*, x, y, \gamma x + (1 - \gamma)y\}$. The uniqueness of the functions u_B and $u_{\{x^*, y^*, x\}}$ implies that $u_B = u_{\{x^*, y^*, x\}}$ on $X_{\{x^*, y^*, x\}}$. Thus $u(x) = u_B(x)$. Similarly, $u(y) = u_B(y)$ and

$$u(\gamma x + (1 - \gamma)y) = u_B(\gamma x + (1 - \gamma)y).$$

As u_B satisfies (22) on X_B , then

$$x \succeq y \implies u(x) = u_B(x) \geq u_B(y) = u(y) \implies u(x) \geq u(y).$$

As u_B is mixture linear, then

$$u(\gamma x + (1 - \gamma)y) = u_B(\gamma x + (1 - \gamma)y) = \gamma u_B(x) + (1 - \gamma)u_B(y) = \gamma u(x) + (1 - \gamma)u(y)$$

and hence, u is mixture linear as well. Thus $u \in \mathcal{U}$ and (22) holds.

Suppose that L is finite dimensional. Then there is a menu A such that the affine hull of A equals the affine hull of X . Index $A = \{a_1, \dots, a_n\}$ so that

$$a_1 \succeq a_2 \succeq \dots \succeq a_n,$$

and let $a_0 = \sum_{i=1}^n \frac{1}{n} a_i$. As X belongs to the affine hull of A , then any $x \in X$ can be written

as an affine combination

$$x = \sum_{i=1}^n \gamma_i a_i \text{ such that } \sum_{i=1}^n \gamma_i = 1.$$

Thus for any $x \in X$, there is $\varepsilon > 0$ such that

$$0 \leq \gamma \leq \varepsilon \implies \gamma x + (1 - \gamma)a_0 \in X_A. \quad (25)$$

As θ is non-null, then there are $x, y \in X$ such that $x \succ y$. Take $\varepsilon > 0$ such that X_A contains both $\varepsilon x + (1 - \varepsilon)a_0$ and $\varepsilon y + (1 - \varepsilon)a_0$. By Independence and (21),

$$x \succ y \implies \varepsilon x + (1 - \varepsilon)a_0 \succ \varepsilon y + (1 - \varepsilon)a_0 \implies a_1 \succ a_n.$$

By Lemma A.3, there is $u \in \mathcal{U}_A$ that satisfies (22) on X_A .

Take $x^* = \frac{a_0 + a_1}{2}$ and $y^* = \frac{a_0 + a_n}{2}$. Take any $z_x, z_y \in X$. By (25), there is $\varepsilon > 0$ such that X_A contains both $\varepsilon z_x + (1 - \varepsilon)x^*$ and $\varepsilon z_y + (1 - \varepsilon)y^*$. As u is non-constant, then by (21), $u(a_1) > u(a_n)$. As u is mixture linear, then $u(x_*) > u(y^*)$ and hence,

$$u(\varepsilon z_x + (1 - \varepsilon)x^*) > u(\varepsilon z_y + (1 - \varepsilon)y^*)$$

for sufficiently small ε . By (22), $\varepsilon z_x + (1 - \varepsilon)x^* \succ \varepsilon z_y + (1 - \varepsilon)y^*$. As $z_x, z_y \in X$ are arbitrary, then θ is stable.

A.3 Proof of Theorem 7

Suppose that $\succeq$ is represented by (15) for some $u \in \mathcal{U}$ and type ψ in L_u . Completeness is obvious. To show transitivity, take any $x, y, z \in X$ such that $x \succeq y \succeq z$. Then $u(x) \geq u(y) \geq u(z)$. If $u(x) > u(z)$, then $x \succeq z$ by (15). If $u(x) = u(z)$, then $x\psi y\psi z$ and hence, $x\psi z$. By (15), $x \succeq z$. Next, θ satisfies Independence because u is mixture linear, and the tie-breaker ψ obeys Independence. Stability has been already derived from representation (4) and a fortiori, from (15).

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